

Solution Requirements

Date	05 July 2026
Team ID	XXXXXX
Project Name	Smart Lender
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>The system registers and identifies users (Credit Officer, Financial Analyst, or Applicant) using name, email, and role profiles to log who processed each loan application.</i>	High
2	Authorization levels	<i>Defines user permissions: Credit Officers can perform predictions and delete log histories, Analysts can view dashboards and historical lists, and Applicants can only submit forms and view their own results.</i>	High

3	External interfaces	<i>Integrates the Flask web backend with local serialized assets: rdf.pkl (containing the tuned XGBoost model and variables) and scale1.pkl (the trained features scaling mapping).</i>	Medium
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4	Transactions processing	<i>Handles form submissions by validating the 11 client features, running inference, calculating confidence scores, and writing records to the local database in a single atomic transaction.</i>	High
5	Reporting	<i>Generates an interactive model comparison list showing accuracies, outputs live EDA visual widgets on the dashboard, and displays historical tables with delete controls.</i>	High
6	Business rules, etc.	<i>All 11 parameters (gender, marital status, income, credit history, etc.) must be mapped to numeric keys. Incomplete files trigger validation warnings, and predictions rely on a tuned XGBoost decision threshold.</i>	High
7	Compliance to laws or regulations	<i>Adheres to local financial data guidelines by securing customer details from unauthorized modifications and ensuring clean records with database cascade deletion.</i>	Medium

8	Other	<i>Implements automatic database cleanup: deleting a prediction log automatically removes the associated applicant profile, credit records, and loan files.</i>	Medium
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Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
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1	Performance & Speed	<i>Response time for page rendering, feature preprocessing, and machine learning classification.</i>	<i>Dashboard pages load in < 1.5 seconds; ML model inference and prediction return in < 200ms.</i>
2	Scalability	<i>Capacity to handle concurrent users and store transactional history logs without performance lags.</i>	<i>Database query response under 50ms; supports up to 1,000 concurrent database connections.</i>
3	Security & Data Privacy	<i>Protection of applicant financial details and safeguarding the backend database.</i>	<i>SQLite database encrypted at rest; query interfaces protected against SQL injections using SQLAlchemy ORM.</i>
4	Reliability & Availability	<i>Application availability and database integrity during processing spikes.</i>	<i>99.9% uptime of the web service; database transactions support auto-rollback on connection failures.</i>

5	Usability & Accessibility	<i>Screen usability, responsive styling layout, and clear indication of evaluation outcomes.</i>	<i>Complies with responsive browser standards (Desktop/Mobile); visual color codes for approval (green) and rejection (red).</i>
6	<i>Other</i>	<i>System maintainability and ease of model retraining.</i>	Modular project architecture separating routing logic, ML training scripts, database schemas, and static styling templates.